

KB-004-Ghost-Availability

KB-004 — Ghost Availability (App Shows Green, Charger Won't Start)

Category: Availability / Telemetry
Call Volume: ~8% of contacts
Impact: Disproportionately high — driver drove to a station expecting to charge
Primary Remote Actions: Real-time status check, status flag, nearest-charger lookup

Why This Matters

Ghost availability — where a charging station appears online and available in an app but fails to deliver power — is small in volume but outsized in its harm to driver trust. Drivers make a deliberate trip to the station based on what the app told them. Arriving to a non-functional charger is one of the most frustrating experiences in the EV ecosystem, especially when time or range is tight. This is also the scenario that predictive fault detection is designed to eliminate upstream.

Background: Why Ghost Availability Happens

Charging station networks rely on OCPP (Open Charge Point Protocol) messages to track the state of each unit. A charger is marked “available” when its last OCPP heartbeat and status notification said it was. The problem: several failure modes can break the charger’s ability to charge without affecting its ability to send heartbeat pings.

Root Causes

Cause	What’s happening	Why app still shows green
Stale telemetry /	Charger went offline but network didn’t	Last status was “Available”; system assumes that’s still true

Cause	What's happening	Why app still shows green
lost OCPP session	receive a clean "offline" message	
Payment terminal fault	Card reader is broken; charger is online but can't authorize sessions	Charger pings network fine; payment module is separate
Firmware hang in authorization module	Charger responds to heartbeat but authorization requests time out	Health check doesn't test authorization path
Network connectivity intermittent	Charger SIM or Ethernet drops; heartbeats get through, session commands don't	Status updates arrive, command responses don't
Power module failure	Charger electronics are on (screen, modem) but power conversion hardware is dead	Control systems operational; power stage failed silently
OCPP version mismatch	A backend update changed the command format; charger can't process new start-session messages	Heartbeats are version-agnostic; session commands aren't
Cable/connector undetected fault	Physical damage not yet triggering a fault status	Self-test hasn't caught the physical failure yet

Diagnostic Script

Step 1 — Confirm the scenario

"I completely understand — it's incredibly frustrating to drive to a station that the app showed as available. Let me check its actual real-time status on our end right now."

Pull real-time station status from the backend operations console (not the driver-facing app status — that may be cached).

Step 2 — Interpret what the backend shows

Backend status	Meaning	Agent action
Available (same as app)	Telemetry lag — charger may be in a failure state we haven't detected yet	Attempt soft reset; if fails, flag as ghost unit
Faulted	Charger has reported a fault, but app cached the last "Available" state	Confirm to driver, apologize, offer nearest alternative
Offline	Charger lost connectivity — OCPP heartbeat stopped	Confirm to driver, flag for connectivity investigation
In session (but driver says no one is charging)	Ghost session — an orphaned session is blocking new starts	Clear the ghost session from the backend, then retry
Reserved	Station was reserved by another driver via app	Explain reservation system, offer nearest unreserved unit

Step 3 — Script for confirmed ghost (backend shows available but charger unresponsive)

"You're right — even though the app showed it as available, this station isn't actually responding to our systems right now. I'm flagging it so it shows correctly for other drivers and gets looked at by our team."

Mark the station with status: "Ghost — availability mismatch." This triggers a tech inspection and a status update push to the app.

Step 4 — Clear a ghost session (if backend shows "in session")

"It looks like there's a stuck session from a previous driver that's preventing new ones from starting. I'm clearing that now — it'll take about 30 seconds."

Clear the orphaned session in the backend. Attempt a soft reset. Ask driver to retry.

Step 5 — Soft reset attempt (for ghost units showing "available")

"I'm going to try a remote restart on the station and see if that brings it back. It'll take about 30 seconds."

Trigger soft reset. If the charger responds and comes back online clean — ask driver to retry starting a session.

If soft reset doesn't restore function:

“The station isn't coming back online through a restart. This will need a technician to look at. I'm really sorry you came out here for this.”

Step 6 — Offer real-time-verified alternative

Do not simply read off the next nearest station from the map — that map may have the same staleness problem.

“Before you head anywhere else, let me verify the status of the nearest alternatives on our end — not just the app, which can lag. Give me just a moment.”

Check backend real-time status for the 2–3 nearest stations. Confirm at least one is genuinely live before recommending.

“I can confirm [Station Name / Address] is showing as live and available on our side right now — not just the app. That's [X miles] from where you are.”

Section A — Explaining the Problem to Drivers Without Jargon

Drivers want to know why the app lied to them. A brief, honest explanation builds more trust than deflection.

Script: > “What happened here is a timing gap between what the station last told our network and what's actually happening right now. Normally that gap is just a few seconds, but if the station loses its connection in a specific way, the last status it sent — ‘available’ — can stick around longer than it should. We're working on making that detection faster, and I've flagged this one so it updates correctly.”

Avoid saying: “the app is wrong” (implies it's always unreliable), “that's just how it works” (dismissive), or “you should have called first” (blaming the driver).

Section B — Ghost Session Clearing

A ghost session is an orphaned OCPP session — a session that was opened (possibly by a previous driver's failed attempt, a firmware crash, or a connectivity loss mid-session) that was never cleanly closed.

Effects: - Station shows “In Session” or “Occupied” to new drivers - New session commands are rejected by the charger - Energy meter may be running on the ghost session, which can generate erroneous charges

To clear: 1. Identify the session ID in the backend 2. Send a remote stop command to that session ID 3. Wait 30 seconds for the charger to confirm the stop 4. Verify the station returns to “Available” in the backend 5. Attempt a soft reset to clear any residual state 6. Ask the driver to try again

If the ghost session has a billing record associated with it, flag it for billing review — it may have generated a charge against a previous driver’s account.

Section C — High-Frequency Ghost Stations

Some stations ghost-out repeatedly — the same unit generates multiple availability mismatch flags per week. These are candidates for:

1. **Firmware upgrade** — a known bug in certain firmware versions causes intermittent heartbeat continuation without functional authorization
2. **SIM / modem replacement** — if the connectivity is intermittent at the hardware level
3. **OCPP configuration review** — incorrect heartbeat interval or status notification settings
4. **Physical inspection** — a power module fault that hasn’t yet triggered a full fault status

Log every ghost-availability call with station ID. If the same station generates 3+ ghost flags in a 7-day period, escalate to Tier 2 for engineering review rather than repeated tech dispatches.

Driver Impact Mitigation

When a driver has made a trip to a non-functional station, consider proactively:

1. **Session credit / goodwill credit** — flag for billing team if the driver was inconvenienced significantly
 2. **Status update subscription** — some apps allow drivers to subscribe to status alerts for a specific station; suggest this if available
 3. **Feedback note** — log the driver’s experience as explicit product feedback; ghost availability is a known systemic issue that engineering teams should be aware of
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Escalation Criteria

Condition	Action
Ghost confirmed, soft reset fails	Flag for tech inspection; offer real-time-verified alternative
Ghost session cleared, but station goes ghost again within the hour	Escalate to Tier 2 engineering
Same station ghosts 3+ times in 7 days	Escalate to Tier 2; remove from recommended alternatives until cleared
Driver drove significant distance and is now stranded	Session credit + Tier 2 escalation
Ghost session had billing implications for another driver	Billing flag for that driver's account

Agent Reminders

- Never recommend a nearest-alternative station without verifying its backend status first. The app map can be stale.
- Always flag ghost stations — a ghost unit that isn't flagged keeps fooling other drivers.
- If a ghost session exists on the backend, clear it before doing a reset — the reset alone may not clear an orphaned session record.
- Ghost availability is a known industry-wide problem; acknowledge it honestly to drivers rather than deflecting.